

5.2) Orthonormalizacija ortogonalnog baze: (v_1, v_2) :

$$\frac{1}{\|v_1\|} = \frac{1}{\sqrt{\int_0^1 1^2 dx}} = \frac{1}{1} \quad \text{normalizacija 1. vektora}$$

$$v = \left\langle \frac{1}{2}, x \right\rangle \frac{1}{\frac{1}{2}} = v = \frac{1}{2} \int_0^1 x^2 dx = x - \frac{1}{2} \int_0^1 1 dx = x - \frac{1}{2} \quad \text{ortogonalizacija 2. vektora na span\{1\} vektora}$$

$$\frac{v \cdot v}{\|v\|^2} = \frac{v \cdot v}{\int_0^1 (x - \frac{1}{2})^2 dx} = \frac{\int_0^1 (x - \frac{1}{2})^2 dx}{\int_0^1 (x^2 - x + \frac{1}{4}) dx} = \frac{\frac{1}{12}}{\frac{1}{12}} = 1 \quad \text{normalizacija 2. vektora}$$

$$x^2 - \left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{\frac{1}{2}} - \left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{\frac{1}{2}} = x^2 - \frac{1}{2} \int_0^1 x^2 dx = x^2 - \frac{1}{6} \int_0^1 1 dx = x^2 - \frac{1}{6} \quad \text{ortogonalizacija 3. vektora na span\{1, 2. vektora\}}$$

$$\frac{x^2 - 6x + \frac{5}{2}}{\|x^2 - 6x + \frac{5}{2}\|} = \frac{x^2 - 6x + \frac{5}{2}}{\sqrt{\int_0^1 (x^2 - 6x + \frac{5}{2})^2 dx}} = \frac{1}{\sqrt{\frac{1}{16} \int_0^1 (x^2 - 6x + \frac{5}{2})^2 dx}} \quad \text{normalizacija 3. vektora}$$

Orthonormalna baza span\{1, x, x^2\} je: $\left\{ \frac{1}{2}, \frac{1}{2}(x - \frac{1}{2}), \frac{1}{\sqrt{16}}(x^2 - 6x + \frac{5}{2}) \right\}$.

Naša. Od projekcije funkcije $\sin x$ na podprostor π baze. OI baze:

$$\left\langle \frac{1}{2}, \sin x \right\rangle = \frac{1}{2} \int_0^1 \sin x dx = \frac{1}{2} \left(-\cos x \right) \Big|_0^1 = \frac{1}{2} (1 - \cos 1) \approx 0.2447$$

Projekcija $\sin x$ na podprostor π baze: $\left(\frac{1}{2}, \frac{1}{2}(x - \frac{1}{2}), \frac{1}{\sqrt{16}}(x^2 - 6x + \frac{5}{2}) \right)$ je: $1.599 - 0.329x - 0.053x^2$.

Specifičnije: računamo normu:

$$\|\sin x - (1.599 - 0.329x - 0.053x^2)\| = \sqrt{\int_0^1 (1.599 - 0.329x - 0.053x^2 - \sin x)^2 dx} \approx 0.349$$

Naša najbolja aproksimacija baze je polinom 0.349 .